

Software Development Life Cycle (SDLC)

NeuroCore Office

1. Project Overview

NeuroCore Office is a Node.js + Express based office document management application focused on PDF management, security, and productivity.

2. Objectives

Provide secure PDF management with upload, search, rename, move, copy, password protection, permission restrictions, and document organization.

3. Technology Stack

Frontend: HTML, CSS, JavaScript

Backend: Node.js, Express.js

Libraries: pdf-lib, QPDF, Multer

Environment: Windows, VS Code/Cursor

4. SDLC Model

Incremental & Iterative development. Each feature is designed, implemented, tested, and integrated before moving to the next feature.

5. Requirements

Functional: Upload PDFs, Search, Rename, Delete, Move, Copy, Protect PDF, Remove Password, Change Password, Permission Restrictions, Document Dashboard.

Non-functional: Performance, Security, Maintainability, Responsive UI.

6. System Design

Client (Browser) ↔ Express Server ↔ File System (uploads/protected/temp) ↔ QPDF/pdf-lib.

7. Implementation Status

Completed: Upload, Search, Rename, Delete, Move, Copy, Document Information, Password Protection, Remove Password, Change Password, Permission Restrictions, Dashboard redesign (in progress).

8. Testing

Unit testing of APIs, integration testing of document workflows, manual UI testing, regression testing after each feature.

9. Deployment

Run using Node.js server. Future deployment can use PM2/Nginx or Docker.

10. Future Enhancements

PDF.js viewer, page thumbnails, merge/split PDFs, OCR, AI-assisted document tools, DOCX/XLSX/PPT support, watermark, digital signature.